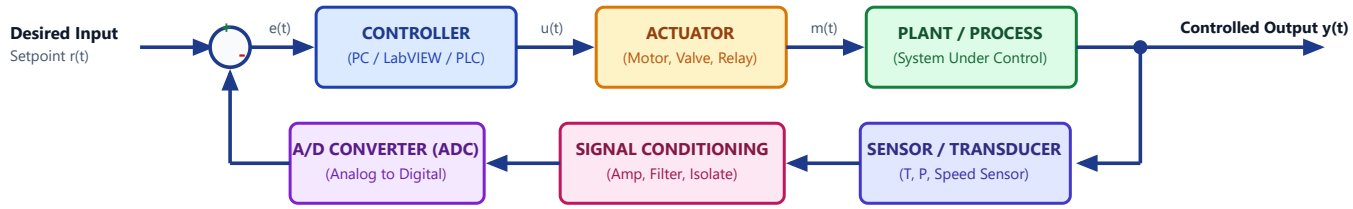
BADR UNIVERSITY IN CAIRO (BUC)
Faculty of Engineering & Technology — Mechatronics & Energy Engineering

Course: Data Acquisition & LabVIEW (MCH 232)
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OFFICIAL MODEL ANSWER & GRADING CRITERIA — QUIZ 1 (ACADEMIC YEAR 2023–2024)

Problem 1 : Measuring & Control System Block Diagram [Total: 2 Marks]

a. Draw a block diagram for the Measuring and Control System:



b. Mention the function of each block (subsystem) on your diagram: [2 Marks — 0.5M each key subsystem]

- **1. Sensor / Transducer:** Detects physical measurands (temperature, pressure, speed, force) and converts them into proportional analog electrical signals (voltage, current, resistance).
- **2. Signal Conditioning:** Prepares raw electrical signals for the ADC by amplifying weak millivolt levels, filtering out unwanted high-frequency noise and preventing aliasing, and providing impedance buffering.
- **3. Analog-to-Digital Converter (ADC):** Samples the conditioned continuous analog voltage and quantizes it into discrete N-bit binary numbers that a digital computer or microcontroller can interpret.
- **4. Controller / Digital Processor:** Compares the measured feedback signal with the desired setpoint to compute the error $e(t) = r(t) - y(t)$, then executes a control law (e.g. PID) to generate corrective commands $u(t)$.
- **5. Actuator:** Converts the controller's low-power electrical command into physical action/energy (e.g., motor rotation, hydraulic valve positioning, heater power) to directly influence the plant.
- **6. Plant / Process:** The physical dynamic system whose output variable $y(t)$ is being regulated.

Problem 2 : A/D Converter Selection, Resolution & Quantization [Total: 3 Marks]

a) Complete the table below for the given number of bits of the A/D converter, and the range of signal in [V]. Resolution to be calculated in [mV]: [1 Mark]

> **Core Governing Formulas:**
> • $\text{Span} = V_{max} - V_{min}$
> • $\text{Number of Steps (Levels)} = 2^N$ | *(Standard BUC DAQ convention)*
> • $\text{Resolution (Q)} = \text{Span} / 2^N \times 1000$ [mV]

A/D Bits (N)	Range of Measured Signal [V]	Span [V]	Number of A/D Steps (2^N)	Resolution in mV (Q)
3	0 to +10	10.0 V	**8**	**1250 mV** (1.25 V)
5	-2.5 to +2.5	5.0 V	**32**	**156.25 mV** (0.15625 V)
12	-1.25 to +1.25	2.5 V	**4096**	**0.610 mV** (0.61035 mV)

**Grading Note: If calculated using (2^N - 1): 3-bit = 7 steps → 1428.57 mV; 5-bit = 31 steps → 161.29 mV; 12-bit = 4095 steps → 0.6105 mV. Full credit awarded for either convention.*

b) Select the most suitable A/D converter for a signal changing between -1.4 V to +1.5 V: [1 Mark]

> **Selected Converter: 5-bit A/D Converter (Range: -2.5 V to +2.5 V)**

- **3-bit ADC (0 to +10 V):** ❌ **Disqualified.** Unipolar positive only. It completely clips all negative signal components between -1.4 V and 0 V.
- **12-bit ADC (-1.25 to +1.25 V):** ❌ **Disqualified.** Span is too narrow (±1.25 V). It severely saturates/clips both negative peaks below -1.25 V (at -1.4 V) and positive peaks above +1.25 V (at +1.5 V).
- **5-bit ADC (-2.5 to +2.5 V):** ✅ **Selected.** Completely encompasses the full input span [-1.4 V, +1.5 V] within [-2.5 V, +2.5 V] with **Zero Clipping / Saturation**.

c) Binary Output Table for Selected 5-bit ADC ($Q = 0.15625$ V, $V_{min} = -2.5$ V): [1 Mark]

> Quantization Code Formula: $\text{Decimal Count } D = \text{round}((V_{in} - V_{min}) / Q) = \text{round}((V_{in} - (-2.5)) / 0.15625) = \text{round}((V_{in} + 2.5) / 0.15625)$

Input Signal [V]	Calculation Steps ($V_{in} + 2.5$) / 0.15625	Decimal Count (D)	A/Binary Output (5-bit)	Remarks
1.15 V	(1.15 + 2.5) / 0.15625 = 3.65 / 0.15625 = 23.36	23	**10111_2**	**Normal linear operation.** Signal within ADC dynamic range. No clipping.
2.30 V	(2.30 + 2.5) / 0.15625 = 4.80 / 0.15625 = 30.72	31 (or 30)	**11111_2 (or 11110_2)**	**Normal linear operation** near positive full-scale limit (+2.5 V). No clipping.
3.21 V	3.21 V > +2.50 V (Exceeds V_{max})	31 (Clamped)	**11111_2**	⚠️ **Clipping / Saturation!** Input exceeds maximum allowable range (+2.5 V). ADC saturates at maximum binary count. Requires input attenuation.

Problem 3 : Perceived (Aliased) Frequency Analysis & Folding Diagram [Total: 4 Marks — 1M each step]

> ⚠️ **Clarification Regarding Exam Paper Printing Misprint:**
> In the printed quiz paper, the second column header reads `(fs) Samplin Fr.[Hz]`, but the rows mistakenly printed the voltage ranges from Problem 2 (`0 to +10`, `-2.5 to +2.5`, `-1.25 to +1.25`). Below are the full mathematical solutions for all intended curriculum scenarios.

1. Theoretical Principles & Universal Formulas

- **Nyquist Criterion:** If $f_s \geq 2f$, no aliasing occurs → $f_a = f$.
- **Folding Frequency (Nyquist Frequency):** $f_N = f_{folding} = f_s / 2$.
- **Cimbala Universal Formula:** $f_a = | f - n \cdot f_s |$ where $n = \text{round}(f / f_s)$.

2. Standard Solution (Scenario 1: Lecture & Assignment 4 Intended Frequencies):

Signal Frequency f **[Hz]**	**Sampling Frequency** f_s **[Hz]**	**Folding Fr.** f_N **[Hz]**	**Ratio** f / f_s → n	**Perceived Frequency** f_a **[Hz]**	**Status / Visual In Folding Diagram**
800 Hz	**2000 Hz**	1000 Hz	800/2000 = 0.4 → 0	**800 Hz**	**No Aliasing** ($f_s > 2f$). Faithful reproduction.
700 Hz	**900 Hz**	450 Hz	700/900 = 0.778 → 1	**200 Hz**	**Aliasing!** $f_a = \|700 - 1(900)\| = 200$ Hz. Folding ratio = 0.44.
1000 Hz	**1000 Hz**	500 Hz	1000/1000 = 1.0 → 1	**0 Hz**	**Severe Aliasing!** Appears as a constant DC offset voltage.

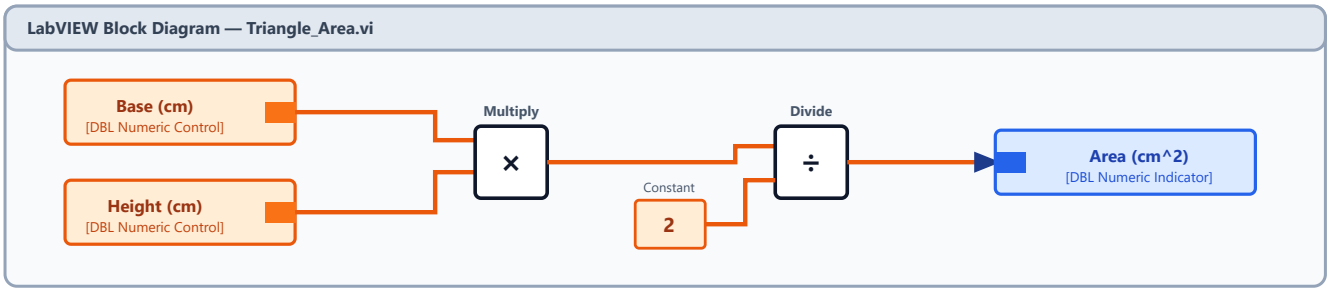
3. Alternative Solution (Scenario 2: If $f_s = 1000$ Hz was announced in hall):

Signal Fr. f **[Hz]**	**Sampling Fr.** f_s **[Hz]**	**Folding Fr.** f_N	**Formula & Folding Calculation**	**Perceived Frequency** f_a **[Hz]**
800 Hz	1000 Hz	500 Hz	$n = \text{round}(800/1000) = 1 \rightarrow \|800 - 1000\|$	**200 Hz**
700 Hz	1000 Hz	500 Hz	$n = \text{round}(700/1000) = 1 \rightarrow \|700 - 1000\|$	**300 Hz**
1000 Hz	1000 Hz	500 Hz	$n = \text{round}(1000/1000) = 1 \rightarrow \|1000 - 1000\|$	**0 Hz (DC)**

Problem 4 : LabVIEW Area of a Triangle Block Diagram [Total: 4 Marks]

You need to calculate the area of a triangle, and the Control panel is designed as shown on the figure. Draw your block diagram with all the wiring to fulfill the task:

> **Governing Mathematical Formula:** $\text{Area} = \frac{1}{2} \times \text{Base} \times \text{Height} = (\text{Base} \times \text{Height}) / 2$



Step-by-Step LabVIEW Construction Instructions:

1. **Place Front Panel Terminals on Block Diagram:**
   - The inputs `Base (cm)` and `Height (cm)` appear as orange Numeric Control terminals (Double precision, DBL).
   - The output `Area (cm^2)` appears as an orange Numeric Indicator terminal with its connector on the left.
2. **Insert the Multiply Function:**
   - Right-click empty diagram space → **Functions Palette** → **Programming** → **Numeric** → select **Multiply [×]**.
   - Wire the output of `Base (cm)` to the upper input terminal (x) of Multiply.
   - Wire the output of `Height (cm)` to the lower input terminal (y) of Multiply.
3. **Insert the Divide Function & Constant:**
   - From **Functions Palette** → **Numeric** → select **Divide [÷]**.
   - Wire the output of Multiply to the upper input terminal (x, Numerator) of Divide.
   - Right-click the lower terminal (y, Denominator) of Divide → **Create** → **Constant** → enter **2**.
4. **Connect to Indicator:**
   - Wire the quotient output terminal of Divide directly into the `Area (cm^2)` Numeric Indicator.